MATH 202 Your name: _____

Learning target A2, version 1

Use definite integrals to find the arc length of a curve.

Consider the curve $y = \cos(x)$ on the interval $[1, 3]$.

(a) Use appropriate graphing technology to help you produce a sketch of this portion of the curve. (As always in calculus, make sure you are in radians mode.)

(b) Give me a reasonable guess for the arc length.

(c) Write down a definite integral that computes the arc length of this curve exactly.

(d) Use appropriate technology to compute this definite integral. (The integrand probably doesn't have an antiderivative!) Report your answer to at least 6 decimal places.

MATH 202 Your name: _____

Learning target A2, version 2

Use definite integrals to find the arc length of a curve.

Consider the curve $y = \sin(x^2)$ on the interval $[0, \sqrt{\pi}]$.

(a) Use appropriate graphing technology to help you produce a sketch of this portion of the curve. (As always in calculus, make sure you are in radians mode.)

(b) Give me a reasonable guess for the arc length.

(c) Write down a definite integral that computes the arc length of this curve exactly.

(d) Use appropriate technology to compute this definite integral. (The integrand probably doesn't have an antiderivative!) Report your answer to at least 6 decimal places.